

Majorana Modes in the Machine

Week 6: From One Measurement to a Phase Diagram

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Week 6 Objective

Goal

Turn the Week 5 proof-of-principle into a diagnostic: sweep μ and ask whether a gate-measurable non-local correlator reconstructs the topological phase boundary.

- Week 5 measured one sweet-spot example at $t = \Delta$, $\mu = 0$.
- Week 6 varies μ across trivial, critical, and topological regimes.
- The target observable remains parity-preserving and non-local.

The Observable We Keep

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$$

- Single-edge operators flip fermion parity, so their expectation vanishes in a parity eigenstate.
- The two-edge string preserves parity and probes the correlation between boundary Majoranas.
- For $L = 4$, this is exactly the Week 5 operator:

$$O_{\text{edge}} = X_0 Z_1 Z_2 X_3.$$

Measurement Protocol

Circuit-level measurement

- Prepare the ground state with the VQE ansatz.
- Rotate qubits with X factors into the Z basis using Hadamards.
- Measure all qubits and multiply bit parities to estimate $\langle O_{\text{edge}} \rangle$.

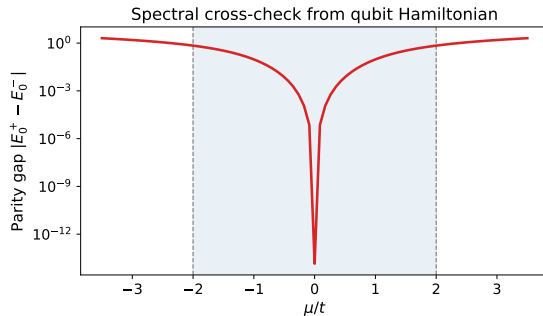
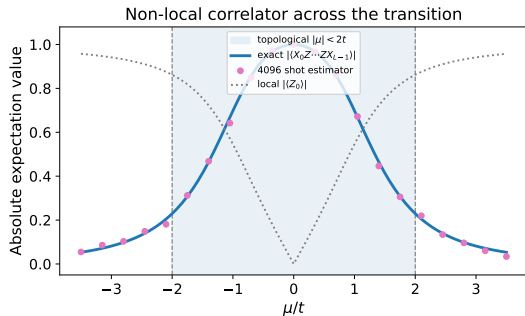
Shot estimator

Each shot returns an eigenvalue $m_s = \pm 1$.

$$\langle O \rangle \approx \frac{1}{N} \sum_{s=1}^N m_s, \quad \sigma \approx \sqrt{\frac{1 - \langle O \rangle^2}{N}}$$

Numerical Phase Sweep

Week 6: Measuring topology by sweeping μ ($L = 4$, $t = \Delta = 1$)



Reading the Plot

- The shaded region marks the known topological interval $|\mu| < 2t$.
- The non-local edge string is large in the topological regime and drops outside it.
- Shot sampling follows the same trend, so the diagnostic is compatible with circuit measurement.
- The parity gap gives an independent qubit-Hamiltonian cross-check of the same phase boundary.

Implementation Step

```
for mu in mu_values:
    H = kitaev_qubit_hamiltonian(L, t, mu, delta)
    psi = even_parity_ground_state(H)
    value = <psi| X0 Z1 ... Z(L-2) X(L-1) |psi>
    shot_value = sample_pm_one(value, shots=4096)
```

- Exact diagonalization is still only a small- L validation path.
- The measured quantity is the same Pauli string a circuit would estimate from bitstrings.
- The next code revision should plug this sweep into the VQE runner directly.

Next: Block 4

- Add a real VQE loop for every μ point instead of ED-prepared validation states.
- Attach depolarizing, readout, and relaxation noise models to the measurement circuits.
- Compare ideal, noisy, and mitigated string-order curves.
- Ask how much noise the topological diagnostic tolerates before the phase boundary is lost.